Physics 9702 Paper 4 · Topic 13

Gravitational fields

The seven hardest A Level Structured Questions on this topic from the 2020 to 2026 papers, hardest first, followed by their mark schemes.

---

1 9702/42 February/March 2021, Question 1 (12 marks) page 2

Examiner report: (b) Many candidates found this calculation challenging. Only the strongest candidates made reference to work needing to be done on the rock in moving it from its orbit to infinity (or vice versa).

2 9702/42 October/November 2023, Question 2 (12 marks) page 4

Examiner report: (b) (i) Most candidates realised that the magnitude of potential energy is obtained by multiplying the potential by the mass, but a significant minority thought that there is a negative relationship between gravitational potential and gravitational potential energy. (e) This was another challenging question.

3 9702/42 October/November 2022, Question 1 (10 marks) page 6

Examiner report: (a) Only a minority of candidates knew the syllabus 'bookwork' that defines gravitational field as force per unit mass. (ii) Only the most able candidates had an understanding that the two forces acting on the object are the weight of the object downwards and the normal contact force upwards from the Earth's surface.

4 9702/41 October/November 2021, Question 2 (9 marks) page 8

Examiner report: A large number of candidates approached this question by discussing forces. The idea that the gravitational field strengths from the Earth and the Moon were both equal in magnitude and opposite in direction was only mentioned by the strongest candidates.

5 9702/42 February/March 2020, Question 1 (10 marks) page 10

Examiner report: (ii) Candidates found this part challenging.

6 9702/42 February/March 2022, Question 1 (10 marks) page 12

Examiner report: (b) Only stronger candidates answered fully correctly but many gave at least one of the points being looked for. It was common to see responses in which gravitational and centripetal forces were discussed as being different forces that somehow balance each other out.

7 9702/41 May/June 2022, Question 1 (10 marks) page 15

Examiner report: (ii) This explanation proved challenging for many candidates, particularly those who referred to forces rather than fields. (iii) This type of calculation can be challenging unless the candidates take the square root of the entire expression in the early stages of the algebra.

Questions were chosen using the Cambridge examiner reports (Principal Examiner Report for Teachers): parts the reports describe as challenging, poorly answered, or answered fully only by the strongest candidates. May/June 2020 and May/June 2026 have no report, so questions from those series were judged directly. Variant 43 repeats variant 41 in every series, so only 41 is cited.

1 **(a)** State Newton's law of gravitation.

.....

.....

..... [2]

(b) Planets have been observed orbiting a star in another solar system. Measurements are made of the orbital radius r and the time period T of each of these planets.

The variation with R^3 of T^2 is shown in Fig. 1.1.

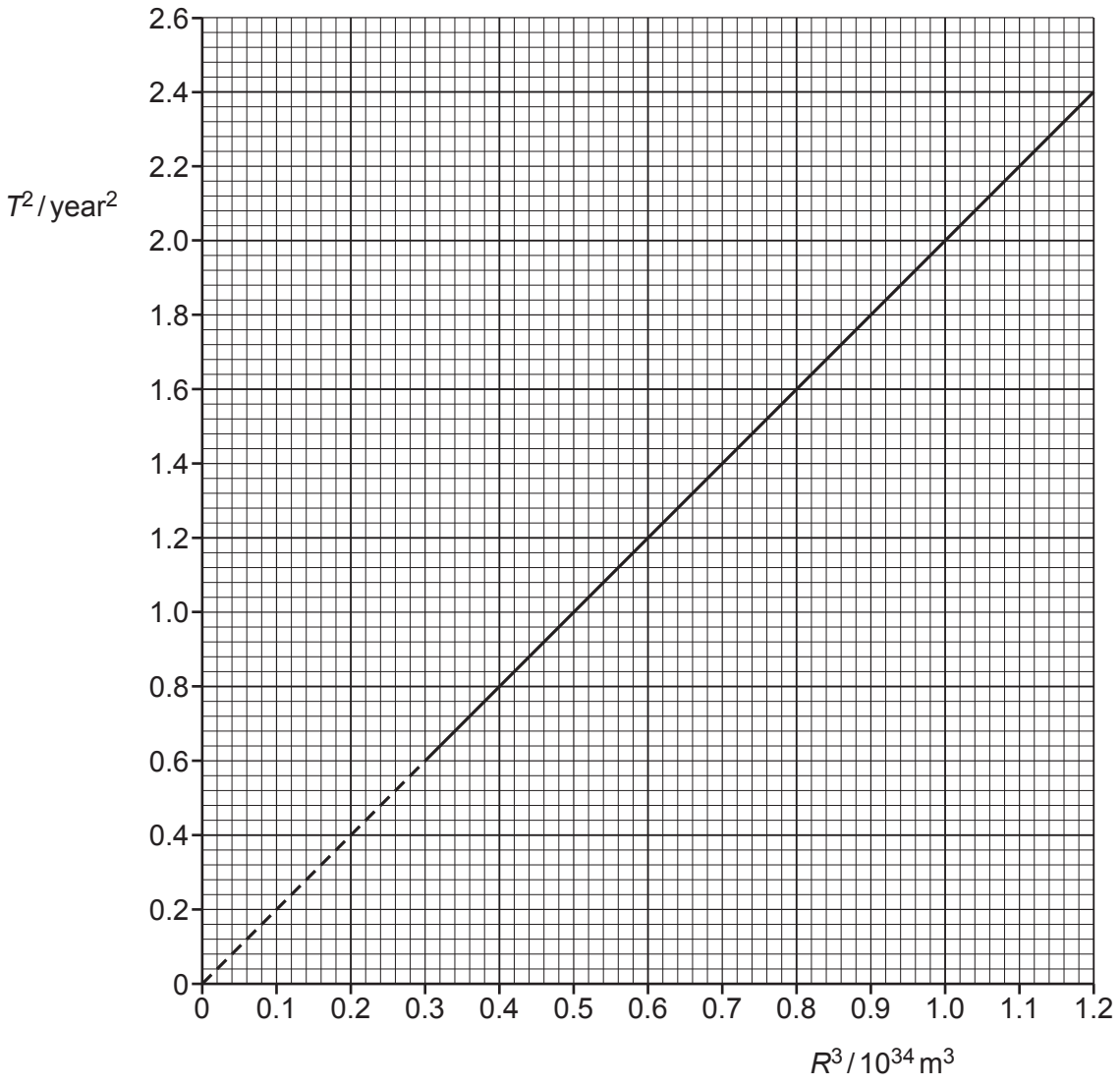


Fig. 1.1

The relationship between T and R is given by

$$T^2 = \frac{4\pi^2 R^3}{GM}$$

where G is the gravitational constant and M is the mass of the star.

Determine the mass M.

M = ... kg [3]

(c) A rock of mass m is also in orbit around the star in **(b)**. The radius of the orbit is r.

(i) Explain why the gravitational potential energy of the rock is negative.

.....

.....

.....

..... [3]

(ii) Show that the kinetic energy E_k of the rock is given by

$$E_k = \frac{GMm}{2r}.$$

[2]

(iii) Use the expression in **(c)(ii)** to derive an expression for the total energy of the rock.

[2]

[Total: 12]

2 **(a)** **(i)** Define gravitational potential at a point.

.....

.....

..... [2]

(ii) The Moon may be considered to be an isolated uniform sphere of mass 7.3×10^{22} kg and radius 1.7×10^{6} m.

Calculate the gravitational potential at the surface of the Moon. Give a unit with your answer.

gravitational potential = ... unit ..... [2]

(b) An isolated uniform spherical planet has gravitational potential ϕ at its surface.

A particle of mass m is projected vertically upwards from the surface. The particle is given just enough kinetic energy to travel to an infinite distance away from the planet, escaping from the gravitational pull of the planet, without any additional work being done on it.

(i) Determine an expression, in terms of m and ϕ, for the gravitational potential energy E_P of the particle at the surface of the planet.

E_P = ... [1]

(ii) Show that the speed v at which the particle is projected upwards from the surface of the planet is given by

$$v = \sqrt{-2\phi}.$$

[2]

(c) A particle is moving upwards at the surface of the Moon.

Use your answer in **(a)(ii)** and the expression in **(b)(ii)** to determine the minimum speed of this particle that will result in it escaping from the gravitational pull of the Moon.

speed = ... $m\,s^{-1}$ [1]

(d) Hydrogen may be assumed to be an ideal gas.
The mass of a hydrogen molecule is 3.34×10^{-27} kg.

Calculate the root-mean-square (r.m.s.) speed of a hydrogen molecule in hydrogen gas that is at a temperature of 400 K.

r.m.s. speed = ... $m\,s^{-1}$ [3]

(e) The surface of the Moon reaches temperatures of approximately 400 K when in direct sunlight.

Use your answers in **(c)** and **(d)** to suggest a reason why the Moon does not have an atmosphere consisting of hydrogen.

.....

..... [1]

[Total: 12]

1 **(a)** Define gravitational field.

.....

..... [1]

(b) A spherical planet can be considered as a point mass at its centre.

(i) On Fig. 1.1, draw gravitational field lines outside the planet to represent the gravitational field due to the planet.

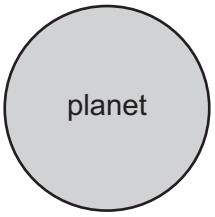


Fig. 1.1

[2]

(ii) A satellite is in a circular orbit around the planet.

Explain, with reference to your answer in **(b)(i)**, why the path of the satellite is circular.

.....

.....

..... [2]

(c) An object rests on the surface of the Earth at the Equator.
The radius of the Earth is 6.4×10^6 m.

(i) Determine the centripetal acceleration of the object.

centripetal acceleration = ... $m\,s^{-2}$ [3]

(ii) Describe how the two forces acting on the object give rise to this centripetal acceleration.
You may draw a diagram if you wish.

.....

.....

..... [2]

[Total: 10]

2 **(a)** Define *gravitational potential*.

.....

.....

..... [2]

(b) The Earth E and the Moon M can both be considered as isolated point masses at their centres. The mass of the Earth is 5.98×10^{24} kg and the mass of the Moon is 7.35×10^{22} kg. The Earth and the Moon are separated by a distance of 3.84×10^{8} m, as shown in Fig. 2.1.

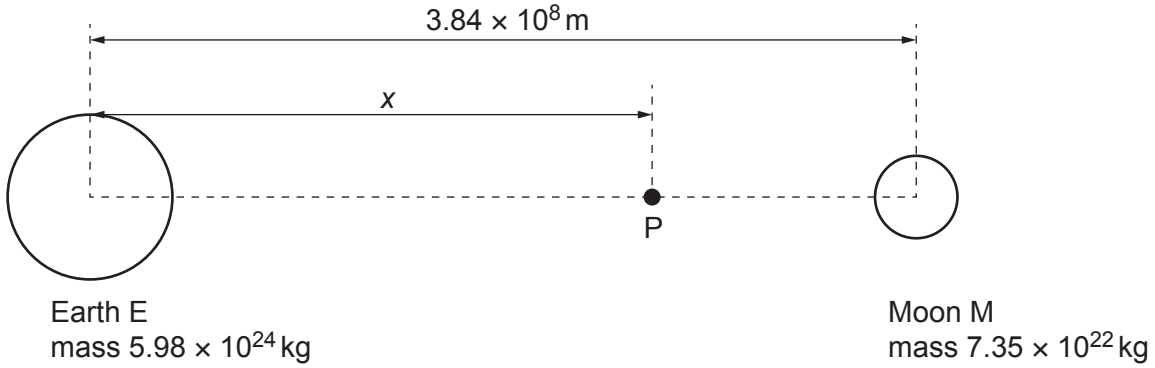


Fig. 2.1 (not to scale)

P is a point, on the line joining the centres of E and M, where the resultant gravitational field strength is zero. Point P is at a distance *x* from the centre of the Earth.

(i) Explain how it is possible for the gravitational field strength to be zero despite the presence of two large masses nearby.

.....

.....

..... [2]

(ii) Show that *x* is approximately 3.5×10^{8} m.

[2]

(iii) Calculate the gravitational potential ϕ at point P.

ϕ = ... $J\,kg^{-1}$ [3]

[Total: 9]

1 **(a)** Define *gravitational potential* at a point.

.....

.....

..... [2]

(b) TESS is a satellite of mass 360 kg in a circular orbit about the Earth as shown in Fig. 1.1.

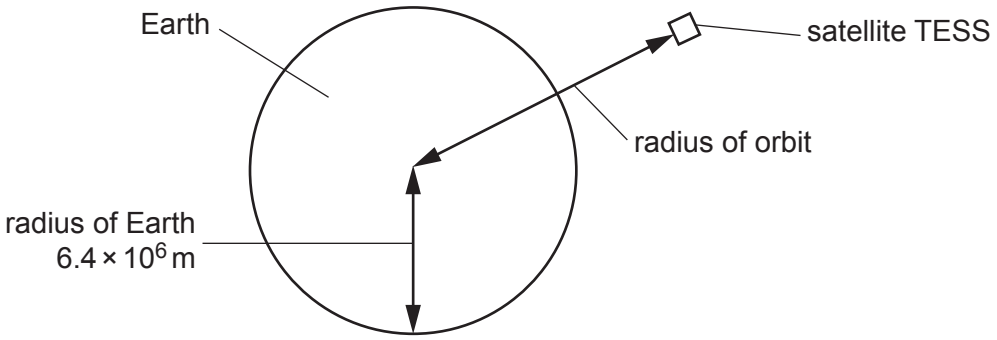


Fig. 1.1 (not to scale)

The radius of the Earth is 6.4×10^6 m and the mass of the Earth, considered to be a point mass at its centre, is 6.0×10^{24} kg.

(i) It takes TESS 13.7 days to orbit the Earth.

Show that the radius of orbit of TESS is 2.4×10^8 m.

[3]

(ii) Calculate the change in gravitational potential energy between TESS in orbit and TESS on a launch pad on the surface of the Earth.

change in gravitational potential energy = ... J [3]

(iii) Use the information in **(b)(i)** to calculate the ratio:

$$\frac{\text{gravitational field strength on surface of Earth}}{\text{gravitational field strength at location of TESS in orbit}}.$$

ratio = ... [2]

[Total: 10]

1 **(a)** The point P in Fig. 1.1 represents a point mass.

On Fig. 1.1, draw lines to represent the gravitational field around P.



Fig. 1.1

[2]

(b) A moon is in circular orbit around a planet.

Explain why the path of the moon is circular.

.....

.....

.....

..... [2]

(c) Many moons are in circular orbit about a planet.

The angular velocity of a moon is ω when the orbit of the moon has a radius r about the planet.

Fig. 1.2 shows the variation of r^3 with $1/\omega^2$ for these moons.

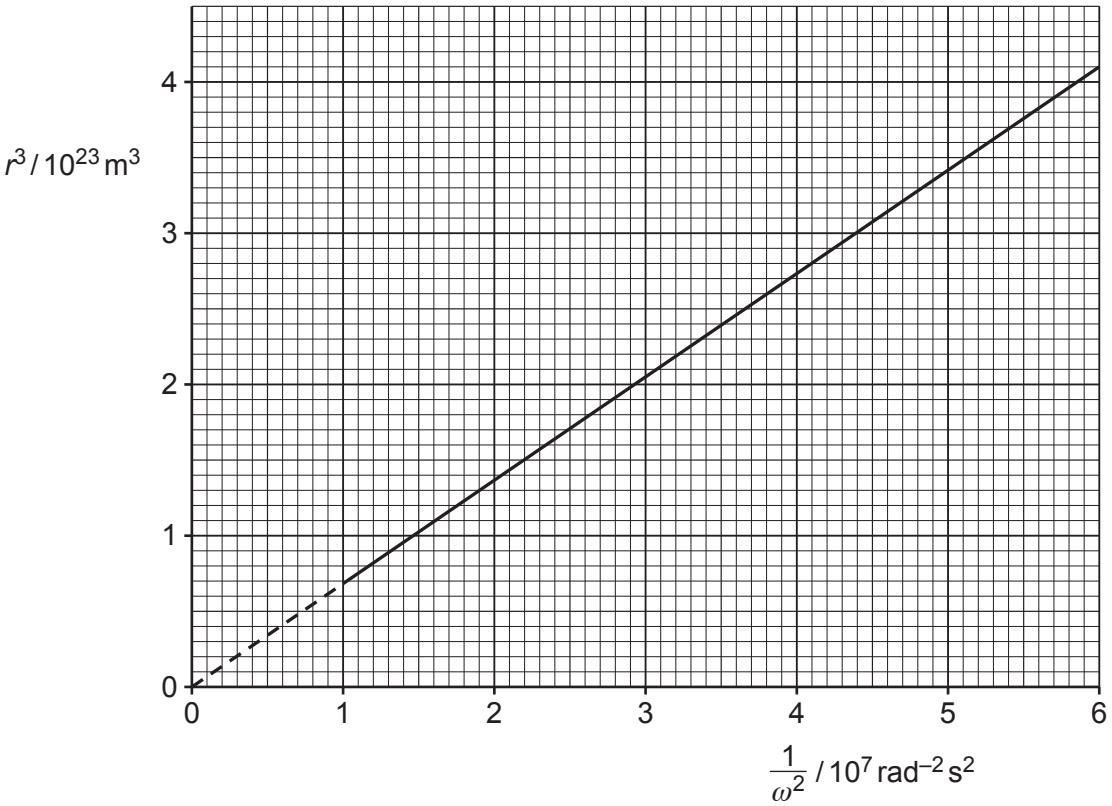


Fig. 1.2

(i) Show that the mass M of the planet is given by the expression

$$M = \frac{\text{gradient}}{G}$$

where G is the gravitational constant.

[2]

(ii) Use Fig. 1.2 and the expression in **(c)(i)** to show that the mass M of the planet is 1.0×10^{26} kg.

[1]

(iii) Determine the speed of a moon in orbit around the planet with an orbital radius of 1.2×10^{8} m.

speed = ... $m\,s^{-1}$ [3]

[Total: 10]

1 **(a)** **(i)** State Newton's law of gravitation.

.....

.....

..... [2]

(ii) Use Newton's law of gravitation to show that the gravitational field strength g at a distance r away from a point mass M is given by

$$g = \frac{GM}{r^2}.$$

[2]

(b) The Earth has a mass of 5.98×10^{24} kg and a radius of 6.37×10^{6} m.
The Moon has a mass of 7.35×10^{22} kg and a radius of 1.74×10^{6} m.
The Earth and the Moon can both be considered as point masses at their centres. Their centres are a distance of 3.84×10^{8} m apart.

(i) Show that the gravitational field strength at the surface of the Moon due to the mass of the Moon is $1.62\,\text{N kg}^{-1}$.

[1]

(ii) Explain why there is a point X on the line between the centres of the Earth and the Moon where the resultant gravitational field strength due to the Earth and the Moon is zero.

.....

.....

..... [2]

(iii) Calculate the distance x of point X from the centre of the Moon.

x = ... m [3]

[Total: 10]

Question	Answer	Marks
1(a)	(gravitational) force is (directly) proportional to product of masses	**B1**
	force (between point masses) is inversely proportional to the square of their separation	**B1**
1(b)	correct read offs from the graph with correct power of ten for R^3	**C1**
	$M = \dfrac{4\times\pi^2\times1.2\times10^{34}}{6.67\times10^{-11}\times2.4\times(365\times24\times3600)^2}$	**C1**
	$=3.0\times10^{30}$ kg	**A1**
1(c)(i)	potential energy is zero at infinity	**B1**
	(gravitational) forces are attractive	**B1**
	work must be done on the rock to move it to infinity	**B1**
1(c)(ii)	$\dfrac{GMm}{r^2}=\dfrac{mv^2}{r}$ *OR* $v^2=\dfrac{GM}{r}$ *OR* $v=\sqrt{\dfrac{GM}{r}}$	**M1**
	use of ½ mv^2 (e.g. multiplication by ½ m) leading to $\dfrac{GMm}{2r}$	**A1**
1(c)(iii)	Ep = ϕm and $\phi = \dfrac{-GM}{r}$ or $E_p=\dfrac{-GMm}{r}$ Total energy = $E_k + E_p$	**C1**
	Total energy $=\dfrac{GMm}{2r}+\dfrac{-GMm}{r}=\dfrac{-GMm}{2r}$	**A1**

Question	Answer	Marks
2(a)(i)	work done per unit mass	**B1**
	work (done) moving mass from infinity (to the point)	**B1**
2(a)(ii)	$\phi = -GM / r$ $= -(6.67 \times 10^{-11} \times 7.3 \times 10^{22}) / (1.7 \times 10^{6})$	**C1**
	$= -2.9 \times 10^{6}\ \text{J kg}^{-1}$	**A1**
2(b)(i)	$E_P = m\phi$	**B1**
2(b)(ii)	$\frac{1}{2}mv^2 + m\phi = 0$	**M1**
	correct algebra leading to $v = \sqrt{(-2\phi)}$	**A1**
2(c)	speed $= \sqrt{(2 \times 2.9 \times 10^{6})}$ $= 2400\ \text{m s}^{-1}$	**A1**
2(d)	$\frac{1}{2}m\langle c^2\rangle = (3/2)kT$	**C1**
	$3.34 \times 10^{-27} \times \langle c^2\rangle = 3 \times 1.38 \times 10^{-23} \times 400$	**C1**
	$c_{\text{r.m.s.}} = 2200\ \text{m s}^{-1}$	**A1**
2(e)	r.m.s. speed is an average so many molecules have speeds greater than the escape speed **or** there is a distribution of molecular speeds (around the r.m.s. value) so many molecules have speeds greater than the escape speed	**B1**

Question	Answer	Marks
1(a)	force per unit mass	**B1**
1(b)(i)	lines drawn are radial from the surface	**B1**
	arrows show pointing towards planet	**B1**
1(b)(ii)	field lines show force (on satellite) is towards centre of planet **or** velocity of satellite is perpendicular to field lines	**B1**
	(gravitational) force perpendicular to velocity causes centripetal acceleration	**B1**
1(c)(i)	T = 24 hours	**C1**
	$a = r\omega^2$ **and** $\omega = 2\pi / T$ **or** $a = v^2 / r$ **and** $v = 2\pi r / T$ **or** $a = 4\pi^2 r / T^2$	**C1**
	$a = (4\pi^2 \times 6.4 \times 10^6) / (24 \times 60 \times 60)^2$ $= 0.034\ \text{m s}^{-2}$	**A1**
1(c)(ii)	identification of the two forces acting on the object as gravitational force and (normal) contact force	**M1**
	gravitational force and normal contact force are in opposite directions, and their resultant causes the (centripetal) acceleration	**A1**

Question	Answer	Marks
2(a)	work done per unit mass	**B1**
	(work done in) moving mass from infinity	**B1**
2(b)(i)	(gravitational) fields from the Earth and Moon are in opposite directions	**B1**
	(resultant is zero where gravitational) fields are equal (in magnitude)	**B1**
2(b)(ii)	$g \propto M / r^2$	**C1**
	$5.98 \times 10^{24} / x^2 = 7.35 \times 10^{22} / (3.84 \times 10^8 - x)^2$ leading to $x = 3.5 \times 10^8$ (m)	**A1**
2(b)(iii)	ϕ (Earth) $= (-)6.67 \times 10^{-11} \times (5.98 \times 10^{24} / 3.5 \times 10^8)$ **and** ϕ (Moon) $= (-)6.67 \times 10^{-11} \times (7.35 \times 10^{22} / 0.38 \times 10^8)$	**C1**
	$\phi = (-)6.67 \times 10^{-11} \times [(5.98 \times 10^{24} / 3.5 \times 10^8) + (7.35 \times 10^{22} / 0.38 \times 10^8)]$	**C1**
	$= -1.3 \times 10^6$ J kg^{-1}	**A1**

Question	Answer	Marks
1(a)	work done per unit mass	**B1**
	work done moving mass from infinity (to the point)	**B1**
1(b)(i)	gravitational force provides centripetal force	**C1**
	$mv^2 / r = GMm / r^2$ and $v = 2\pi r / T$ OR $mr\omega^2 = GMm / r^2$ and $\omega = 2\pi / T$ OR $r^3 = GMT^2 / 4\pi^2$	**C1**
	$r^3 = 6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times (13.7 \times 24 \times 3600)^2 / 4\ \pi^2$ so r = 2.4×10^8 m	**A1**
1(b)(ii)	(E_P = –) GMm / r work done = $GMm / r_1 - GMm / r_2$	**C1**
	= $6.67 \times 10^{-11} \times 360 \times 6.0 \times 10^{24}$ $(1/6.4 \times 10^6 - 1 / 2.4 \times 10^8)$	**C1**
	= **2.2×10^{10}** J	**A1**
1(b)(iii)	$g = GM / r^2$	**C1**
	ratio = r_{TESS}^2 / r_{earth}^2 = $(2.4 \times 10^8 / 6.4 \times 10^6)^2$ = **1400**	**A1**

Question	Answer	Marks
1(a)	at least 4 straight radial lines to P 	**B1**
	all arrows pointing along the lines towards P	**B1**
1(b)	**Any 2 from:** gravitational force provides the centripetal force (centripetal or gravitational) force has constant magnitude (centripetal or gravitational) force is perpendicular to velocity (of moon) / direction of motion (of moon)	**B2**
1(c)(i)	$\frac{GMm}{r^2} = mr\omega^2$	**M1**
	$M = \frac{r^3\omega^2}{G}$ and gradient = $r^3\omega^2$ hence $M = \frac{\text{gradient}}{G}$ or $r^3 = GM \times 1/\omega^2$ so gradient = GM hence $M = \frac{\text{gradient}}{G}$	**A1**
1(c)(ii)	$M = 4.1 \times 10^{23} / (6.0 \times 10^7 \times 6.67 \times 10^{-11}) =$ **1.0×10^{26}** kg	**B1**

Question	Answer	Marks
1(c)(iii)	$\frac{GMm}{r^2} = \frac{mv^2}{r}$ $\frac{GM}{r} = v^2$	**C1**
	$v^2 = \frac{6.67 \times 10^{-11} \times 1.0 \times 10^{26}}{1.2 \times 10^{8}}$ $v^2 = 5.6 \times 10^{7}\ \text{m s}^{-1}$	**C1**
	$v = 7500\ \text{m s}^{-1}$	**A1**

Question	Answer	Marks
1(a)(i)	(gravitational) force is (directly) proportional to product of masses	**B1**
	force (between point masses) is inversely proportional to the square of their separation	**B1**
1(a)(ii)	$g = F / m$	**C1**
	$F = GMm / r^2$ and so $g = [GMm / r^2] / m = GM / r^2$	**A1**
1(b)(i)	$g = (6.67 \times 10^{-11} \times 7.35 \times 10^{22}) / (1.74 \times 10^{6})^2 = 1.62\ \text{N kg}^{-1}$	**A1**
1(b)(ii)	fields (due to Earth and the Moon) have equal magnitudes	**B1**
	fields (due to Earth and the Moon) are in opposite directions	**B1**
1(b)(iii)	distance of X from Earth = $(3.84 \times 10^{8} - x)$	**C1**
	$(G \times)\ 7.35 \times 10^{22} / x^2 = (G \times)\ 5.98 \times 10^{24} / (3.84 \times 10^{8} - x)^2$	**C1**
	$x = 3.8 \times 10^{7}$ m	**A1**